COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

CEP LAB

Lab Title: Traffic Management System

1. OBJECTIVE

The purpose of this CEP lab is to design and implement a C++ console program that solves a real traffic management problem. The program must allow a user to enter vehicle counts for four roads, classify the traffic density of each road as Low, Moderate, or Heavy, and automatically calculate green signal timing for each road based on the number of vehicles. The program uses a class to represent each road, an array of class objects, and separate functions for traffic analysis and signal timing.

2. THEORY

Traffic signal timing systems are an essential part of smart urban traffic management. In a manual or automated signal system, the duration of a green light is directly related to the volume of traffic on a given road. A road with fewer vehicles requires a shorter green phase, while a heavily congested road needs a longer green phase to clear the queue efficiently.

Traffic density classification used in this program:

- Low Traffic: fewer than 20 vehicles — Green Signal: 20 seconds
- Moderate Traffic: 20 to 50 vehicles — Green Signal: 40 seconds
- Heavy Traffic: more than 50 vehicles — Green Signal: 60 seconds

C++ concepts applied in this program:

- Class with public data members and member functions
- Array of class objects to store data for four roads
- Functions outside the class for traffic analysis and signal timing
- do-while loop with switch-case for a menu-driven interface

3. SOFTWARE USED

- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)

4. PROCEDURE

Task — Traffic Management System

Write a C++ program that takes vehicle counts from four roads, analyzes the traffic density of each road, and assigns green signal timing accordingly. The program uses a class called Road with data members for road name and vehicle count, along with member functions for input and display. Two standalone functions handle traffic status classification and signal timing calculation. A menu-driven do-while loop allows the user to enter data, display it, check traffic status, and view signal timing.

```
//  NAME: Abdul Ghafar
//  REGISTRATION NO: BF25NWELE0711
//  SECTION: A (POWER)
//  DEPARTMENT: ELECTRICAL ENGINEERING
//  SEMESTER: 2ND (SPRING)
// TRAFFIC MANAGEMENT SYSTEM
// This program takes vehicle counts from 4 roads,
// analyzes traffic density, and assigns signal timing.
// ==================================================
#include <iostream>     // For input and output
#include <string>       // For string handling
using namespace std;

// Class to represent a road
class Road
{
public:
    string roadName;    // Name of the road
    int vehicles;       // Number of vehicles on the road

    // Function to input road data
    void inputData()
    {
        cout << "Enter Road Name: ";
        cin >> roadName;

        cout << "Enter Number of Vehicles: ";
        cin >> vehicles;
    }

    // Function to display road data
    void displayData()
    {
        cout << "\nRoad Name: " << roadName;
        cout << "\nVehicles: " << vehicles << endl;
    }
};

// Function to determine traffic condition
void trafficStatus(int vehicles)
{
    if (vehicles < 20)
    {
        cout << "Traffic Status: LOW\n";
    }
    else if (vehicles >= 20 && vehicles <= 50)
    {
```

```cpp
        cout << "Traffic Status: MODERATE\n";
    }
    else
    {
        cout << "Traffic Status: HEAVY\n";
    }
}

// Function to calculate signal timing
void signalTiming(int vehicles)
{
    cout << "\nSignal Timing Information\n";

    if (vehicles < 20)
    {
        cout << "Green Signal Time: 20 seconds\n";
    }
    else if (vehicles >= 20 && vehicles <= 50)
    {
        cout << "Green Signal Time: 40 seconds\n";
    }
    else
    {
        cout << "Green Signal Time: 60 seconds\n";
    }
}

// Main function
int main()
{
    // Array of roads
    Road roads[4];

    // Variable for menu choice
    int choice;

    do
    {
        cout << "\n================================";
        cout << "\n   TRAFFIC MANAGEMENT SYSTEM";
        cout << "\n================================";
        cout << "\n1. Enter Traffic Data";
        cout << "\n2. Display Traffic Data";
        cout << "\n3. Check Traffic Status";
        cout << "\n4. Calculate Signal Timing";
        cout << "\n5. Exit";
        cout << "\nEnter Choice: ";
        cin >> choice;

        switch(choice)
        {
        case 1:

            // Input data for all roads
            for(int i = 0; i < 4; i++)
            {
                cout << "\nRoad " << i + 1 << endl;
                roads[i].inputData();
            }
            break;

        case 2:

            // Display data of all roads
            cout << "\n------ Traffic Data ------\n";
```

```cpp
            for(int i = 0; i < 4; i++)
            {
                roads[i].displayData();
            }
            break;

        case 3:

            // Check traffic status
            cout << "\n------ Traffic Analysis ------\n";

            for(int i = 0; i < 4; i++)
            {
                cout << "\nRoad: " << roads[i].roadName << endl;

                trafficStatus(roads[i].vehicles);
            }
            break;

        case 4:

            // Display signal timings
            cout << "\n------ Signal Timing ------\n";

            for(int i = 0; i < 4; i++)
            {
                cout << "\nRoad: " << roads[i].roadName << endl;

                signalTiming(roads[i].vehicles);
            }
            break;

        case 5:

            cout << "\nExiting Program...";
            break;

        default:

            cout << "\nInvalid Choice! Try Again.";
        }

    } while(choice != 5);

    return 0;   // Program ends successfully
}
```

Output:

```
================================
   TRAFFIC MANAGEMENT SYSTEM
================================
1. Enter Traffic Data
2. Display Traffic Data
3. Check Traffic Status
4. Calculate Signal Timing
5. Exit
Enter Choice: 1

Road 1
Enter Road Name: Main_Bazaar
Enter Number of Vehicles: 10

Road 2
Enter Road Name: GT_Road
Enter Number of Vehicles: 35

Road 3
Enter Road Name: Ring_Road
Enter Number of Vehicles: 70

Road 4
Enter Road Name: School_Road
Enter Number of Vehicles: 25

================================
   TRAFFIC MANAGEMENT SYSTEM
================================
1. Enter Traffic Data
2. Display Traffic Data
3. Check Traffic Status
4. Calculate Signal Timing
5. Exit
Enter Choice: 3

------ Traffic Analysis ------

Road: Main_Bazaar
Traffic Status: LOW

Road: GT_Road
Traffic Status: MODERATE

Road: Ring_Road
Traffic Status: HEAVY

Road: School_Road
Traffic Status: MODERATE

Enter Choice: 4

------ Signal Timing ------

Road: Main_Bazaar
Signal Timing Information
```

```
Green Signal Time: 20 seconds

Road: GT_Road
Signal Timing Information
Green Signal Time: 40 seconds

Road: Ring_Road
Signal Timing Information
Green Signal Time: 60 seconds

Road: School_Road
Signal Timing Information
Green Signal Time: 40 seconds

Enter Choice: 5

Exiting Program...
```

5. CONCLUSION

This CEP lab demonstrated the use of multiple C++ programming concepts in a single practical program. A class was used to group the road name and vehicle count into one unit, with member functions for data input and display. An array of class objects stored data for all four roads. Separate standalone functions handled traffic density classification using an if-else structure and green signal timing calculation based on vehicle count thresholds. A do-while loop with a switch-case statement provided a menu-driven interface for the user. The program correctly classified traffic conditions and assigned appropriate signal durations for all test inputs, successfully combining object-oriented concepts with practical traffic engineering logic.